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A

Project Synopsis

on

AI Memory Reconstruction from Digital Footprint

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Abstract

Modern digital life produces enormous amounts of data across devices and platforms —photos, chats, social media posts, emails, browser activity, and location traces. Humans forget experiences, but their digital footprint silently records them. Manual organization of these data streams is nearly impossible, leading to memory loss, difficulty retrieving past events, and fragmented personal history.

This project proposes an **AI-driven memory reconstruction system** that automatically ingests multimodal digital data, preprocesses and extracts meaningful signals (text, images, metadata), and uses modern multimodal embeddings, vector databases, and retrieval-augmented generation (RAG) to recreate structured, narratively coherent "memories." The system builds personal timelines, summarizes key events, clusters episodes, and generates natural-language recollections grounded strictly in the user's verified data.

The proposed system includes modules for data ingestion, preprocessing, embedding generation, similarity search, LLM-based narrative reconstruction, and a user-friendly memory timeline interface. Expected outcomes include a working prototype with secure on-device or cloud-optional processing, customizable memory generation, and high factual accuracy through RAGbased grounding. Performance targets include <1s retrieval latency for top-k memory units and >90% factual alignment with source data as evaluated by human users.

1. Introduction

1.1 Brief Project Overview

This project develops an AI-powered software pipeline that captures a user's digital footprint, converts it into standardized event objects, and reconstructs personal memories using multimodal AI techniques. The pipeline includes data cleaning, OCR/image analysis, metadata extraction, embedding computation, vector indexing, event similarity clustering, and memory synthesis using large language models (LLMs). A front-end dashboard provides an interactive, chronological visualization of reconstructed memories, enabling users to revisit past experiences with clarity and accuracy.

1.2 Motivation

People forget events easily, but smartphones, social platforms, and digital ecosystems store thousands of traces of daily life. No existing system reconstructs life events **automatically**, **accurately**, and **privately**. Users need a reliable memory assistant that can:

- Recreate past events from multimedia data
- Summarize days, weeks, or phases of life
- Provide searchable personal history
- Organize thousands of photos, messages, and interactions meaningfully

This project addresses the rising cognitive load of digital life, where personal data is fragmented across apps and platforms, making recollection difficult.

1.3 Gaps in Existing Work

- Digital assistants (Google Photos, iOS Memories, Facebook Memories) provide isolated highlights—not full memory reconstruction.

- No existing unified pipeline integrates **multimodal embeddings + vector retrieval + RAG** for personal memory generation.
- Most tools focus on either photos or text—not the entire multi-source digital footprint.
- Very few systems allow **complete user-controlled privacy**, on-device processing, or source-verified memory explanations.

1.4 Contribution of This Project

1. A modular pipeline for ingesting multimodal digital data (text, images, metadata, device logs).
2. A multimodal embedding and vector search engine enabling event similarity and clustering.
3. A RAG-based narrative generator that produces factual, evidencegrounded memory summaries.
4. A dashboard visualizing timelines, clusters, and memory cards for intuitive exploration.
5. A privacy-first architecture supporting optional offline/on-device memory reconstruction.

1.5 Technology Used and Field

Field: Artificial Intelligence, Deep Learning, Personal Data Modeling, Natural Language Processing.

Technologies: Python, FastAPI, React.js, PostgreSQL, FAISS/Pinecone, CLIP/Sentence Transformers, Whisper, Tesseract OCR, LLMs (RAG-based), embedding models.

Chapter 02

Literature Survey

Digital Memory Systems

Projects like Microsoft MyLifeBits demonstrated how digital lifelogging can support memory, but lacked AI-driven reconstruction capabilities. Current cloud services primarily focus on photo-based highlights.

Multimodal Embeddings

Models like CLIP (Radford et al., 2021) and Sentence Transformers enable unified embeddings across text and images, essential for event similarity matching.

Retrieval-Augmented Generation (RAG)

RAG (Lewis et al., 2020) grounds LLM outputs in factual retrieved data, reducing hallucinations and improving accuracy in personal memory reconstruction.

Personal Knowledge Modeling

Tools like Notion and Roam Research emphasize manual input; automation remains limited.

Gaps Analysis

- No automated, private, end-to-end memory reconstruction system exists.
- Lack of research on combining OCR, embeddings, vector DB, and LLMs for personal-scale data.

Fragmentation across devices creates difficulty for meaningful personalization.

Chapter 03

Methodology / Planning of Work

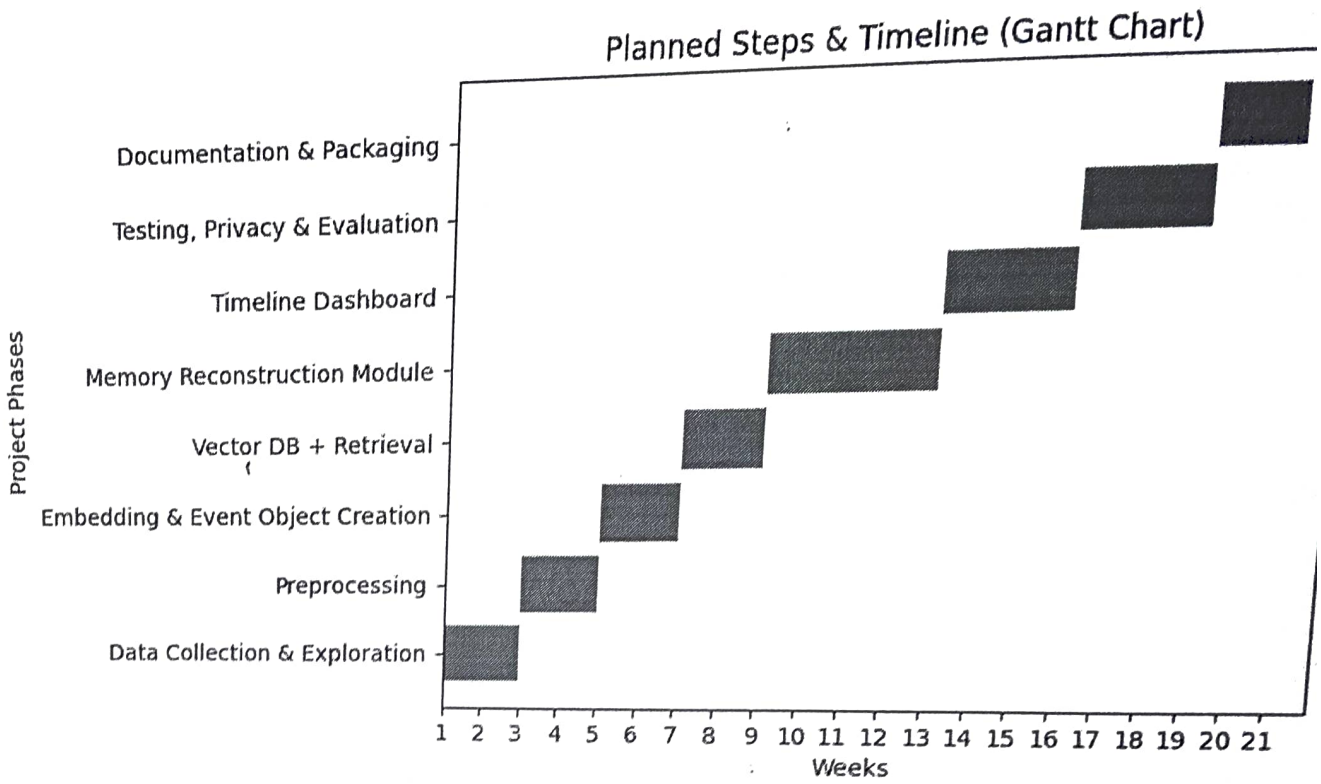


Figure 1: Gantt Chart Showing Planned Steps and Timeline

Planned Steps & Timeline

The proposed system will be implemented over a total duration of **20 weeks**, following a phased and systematic development approach to ensure timely completion and quality outcomes.

During **Weeks 1–2**, the project will focus on **Data Collection and Exploration**. Sample digital footprints such as photographs, chat logs, and location histories will be collected and analyzed to understand data formats, structure, and variability.

In **Weeks 3–4**, **Preprocessing** of the collected data will be carried out. This phase includes Optical Character Recognition (OCR) for images, extraction of EXIF metadata, timestamp normalization, parsing of chat data, and standardization of metadata to ensure consistency across heterogeneous sources.

The **Embedding and Event Object Creation** phase will be conducted during **Weeks 5–6**. In this phase, vector embeddings will be generated for textual and visual data using

suitable embedding models. The processed data will be transformed into unified "event objects" representing meaningful memory units.

During **Weeks 7–8**, a **Vector Database and Retrieval Module** will be implemented using FAISS or Pinecone to enable efficient similarity-based retrieval of stored event embeddings.

The **Memory Reconstruction Module** will be developed in **Weeks 9–12** using a Retrieval-Augmented Generation (RAG) approach. This module will generate coherent and factually grounded narratives from retrieved events while minimizing hallucination and maintaining contextual accuracy.

In **Weeks 13–15**, a **Timeline Dashboard** will be designed and implemented. This user interface will allow users to browse reconstructed memories chronologically along with verified supporting evidence.

The **Testing, Privacy, and Evaluation** phase will be conducted during **Weeks 16–18**. System accuracy, latency, and factual alignment will be evaluated, and privacy-preserving measures will be verified.

Finally, **Documentation and Packaging** will be completed in **Weeks 19–20**, including preparation of the final project report, deployment documentation, and a demonstration video.

Chapter 04

Requirements

Software Requirements

- Python 3.10+
- PostgreSQL
- NumPy
- Pandas
- Scikit-learn

Tools / Frameworks Requirements

- FastAPI
- FAISS
- Pinecone
- Sentence-Transformers
- CLIP
- Whisper
- Tesseract OCR
- React.js
- Docker

Hardware Requirements

- Development laptop (16 GB RAM, 8-core CPU)
- GPU recommended for embedding generation
- 500 GB storage for datasets

References (IEEE Format)

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